Allison Sullivan

Associate Professor, Director UG Software Engineering Program
Department of Computer Science and Engineering
The University of Texas at Arlington

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https://allisonius.github.io/
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RESEARCH INTEREST

Automated Software Engineering: Test Generation, Fault Localization, Automated Program Repair and Mutation Testing

Formal Methods and Programming Languages: Specification Correctness, Quality Metrics for Specifications, Maintainability of Specifications, and Scenario Refinement/Generation

Human Computer Interaction: Live Programming, Bidirectional Program Transformations, and Usability of Specifications/Tools

ACADEMIC EMPLOYMENT HISTORY

2026 – PRESENT **Associate Professor** The University of Texas at Arlington **Arlington, TX**

2026 – PRESENT **Faculty Fellow, Honor's College** The University of Texas at Arlington **Arlington, TX**

2025 – PRESENT **Director of the UG Software Engineering Program** The University of Texas at Arlington **Arlington, TX**

2020 – 2026 **Assistant Professor** The University of Texas at Arlington **Arlington, TX**

2018 – 2020 **Assistant Professor,** North Carolina A&T State University **Greensboro, NC**

EDUCATION

2014-2017 **Ph.D. in Software Engineering,** The University of Texas at Austin **Austin, TX**

2012-2014 **M.S. in Software Engineering,** The University of Texas at Austin **Austin, TX**

2008-2012 **B.S. in Software Engineering,** The University of Texas at Dallas **Richardson, TX**

INDUSTRY EMPLOYMENT HISTORY

June – July 2019 **Google, Inc,** Faculty in Residence (FIR) **Mountain View, CA**
FIR is an immersive program that trained me to: design and implement classroom experiments related to project-based learning, conduct resume workshops, and give mock interviews.

2015 – 2016 **Naval Research Laboratory,** Intern in the Center for High Assurance Computer Systems **Washington, D.C**
I extended the SCR toolset to automatically generate test cases that incorporate assumptions. This included updating the automated translations of SCR models into SPIN, NuSMV, and PVS.

May – Aug. 2013 **IBM Research ,** Intern in the Test Suite Automation Team **Austin, TX**
I built a test automation toolset to ensue test suites automatically feed their output into RQM, IBM's project management tool. This toolset bridges existing automation scripts with the RQM and is in place at IBM today.

HONORS AND AWARDS

2025 UTA's Blavatnik National Awards for Young Scientists nominee for Physical Sciences & Engineering Category
2025 Arlington's Chamber of Commerce 40 under 40
2025 Dagstuhl Seminar Organizer
2025 UTA's College of Engineering Outstanding Early Career Research Award
2024 UTA's CSE Department Rising Star Research Award
2024 NSF CISE CAREER Workshop – Awardee Speaker
2024 NSF CAREER Award
2019 NSF HBCU-UP ACE Implementation Project – "*Project Based Learning in the Teaching of Analysis of Algorithms*"
2019 Grace Hopper Faculty Scholar
2019 2nd Place Overall at SAE International AutoDrive Challenge (Faculty Mentor, Functional Safety, NC A&T)

RESEARCH FUNDING

7 grants | **Total amount:** $10,084,000 | **My share:** $2,324,000

4 Grants Obtained at UTA | **My share:** $1,534,000

[1] Model-By-Numbers: An Interactive Educational Tool for Software Models
National Science Foundation IUSE ESL, solo PI, $400k, Oct 2025 - Sept 2028

[2] CAREER: Live Programming for Finite Model Finders
National Science Foundation, solo PI, $525k, June 2024 - May 2029
REU Supplement, $20k, June 2026

[3] SHF: Small: INCA: Incremental Analysis of Software Specification for Evolving Systems
National Science Foundation – CISE Core, solo PI, $490k, Oct 2022 - Sept 2025

[4] KeenEye: Enhancing Scenario Enumeration
National Science Foundation – Formal Methods in the Field, solo PI, $99k, Oct 2021 - Mar 2023

3 Grants Obtained at NC A&T | **My share:** $790,000

[5] MRI: Acquisition of a Testbed of Connected Autonomous MicroTransit Vehicles
National Science Foundation Major Research Instrumentation, Co-PI, $550k (My Share: $55k), July 2020 - June 2022

[6] Secure and Safe Assured Autonomy (S^2A^2)
NASA University Leadership Initiative, Co-PI, $7.9M (My Share: $635k), Aug 2020 - Aug 2024

[7] Alloy Analyzer Plus: An Integrated Development Environment for Alloy
National Science Foundation – Formal Methods in the Field, solo PI, $100k, Transferred to UTA July 2020

REFEREED JOURNAL PUBLICATIONS

(J2) Ana Jovanovic, Allison Sullivan. Mutation Testing for Temporal Alloy Models (Extended Version) In *Software and Systems Modeling* (SoSyM), Dec. 2025. (Impact Factor: 2.1)

(J1) Manfred Broy, Achim D. Brucker, Alessandro Fantechi, Mario Gleirscher, Klaus Havelund, Markus Kuppe, Alexandra Mendes, André Platzer, Jan Oliver Ringert, Allison Sullivan. Does Every Computer Scientist Need to Know Formal Methods? In *Formal Aspects of Computing* (FAOC), April. 2025. (Impact Factor: 1.0)

REFEREED CONFERENCE PUBLICATIONS

(C27) Mohammad Patwary, Allison Sullivan. Formula Completion Suggestions for Alloy Models with Selection Guidance. In *the 41st IEEE/ACM International Conference on Automated Software Engineering* (ASE 2026), To Appear. (Acceptance: 21.5%)

(C26) Mohammad Patwary, Allison Sullivan. Towards Output Directed Debugging of Finite Model Finders. In *The ACM International Conference on the Foundations of Software Engineering - Ideas, Visions and Reflections Track* (FSE IVR 2026), July 2026. (Acceptance: 37.5%)

(C25) Ana Jovanovic, Mohammad Patwary, Allison Sullivan. Refining Alloy-Based Mutation Operators to Reflect Common Mistakes. In *Proceedings of the 13th International Conference on Formal Methods in Software Engineering* (FormaliSE 2025), pages 1–11, June 2025. (Acceptance: 31%) ***Invited for Journal Submission.***

(C24) Ana Jovanovic, Allison Sullivan. Right or Wrong: Understanding How Novice Users Write Software Models. In *the 22nd International Conference on Software Engineering and Formal Methods* (SEFM 2024), pages 309–327. Nov. 2024. (Acceptance: 33.8%)

(C23) Guanxuan Wu, Allison Sullivan. AlloyASG: Alloy Predicate Code Representation as a Compact Structurally Balanced Graph. In *Proceedings of the ACM/IEEE 27th International Conference on Model-Driven Engineering Languages and Systems* (MODELS 2024), pages 57–68, Sept. 2024. (Acceptance: 23.5%)

(C22) Adam Emerson, Allison Sullivan. Crucible: Graphical Test Cases for Alloy Models. In *Proceedings of the 34th International Symposium on Software Reliability Engineering* (ISSRE 2023), pages 218-227, Oct. 2023. (Acceptance: 29.5%)

(C21) Ana Jovanovic, Allison Sullivan. Mutation Testing for Temporal Alloy Models. In *Proceedings of the ACM/IEEE 26th International Conference on Model-Driven Engineering Languages and Systems* (MODELS 2023), pages 228-238, Oct. 2023. (Acceptance: 24.6%) ***Invited for Journal Submission.***

(C20) Allison Sullivan. Integrating Testing into the Alloy Model Development Workflow. In *Proceedings of the ACM/IEEE 26th International Conference on Model-Driven Engineering Languages and Systems* (MODELS 2023), pages 117-128, Oct. 2023. (Acceptance: 24.6%)

(C19) Allison Sullivan. Live Programming for Finite Model Finders. In *Proceedings of the 38th IEEE/ACM International Conference on Automated Software Engineering - New Ideas and Emerging Results* (ASE-NIER 2023), pages 1747-1752, Sept. 2023. (Acceptance: 36%)

(C18) Jan Oliver Ringert, Allison Sullivan. Abstract Alloy Instances. In *Proceedings of the 25th International Symposium on Formal Methods* (FM 2023), pages 364-382, Mar. 2023.

(C17) Ana Jovanovic, Allison Sullivan. REACH: Refining Alloy Scenarios by Size. In *Proceedings of the 33rd International Symposium on Software Reliability Engineering* (ISSRE 2022), pages 229–238, Oct. 2022. (Acceptance: 29%)

(C16) Ana Jovanovic, Allison Sullivan. Towards Automated Input Generation for Sketching Alloy Models. In *Proceedings of the 10th International Conference on Formal Methods in Software Engineering* (FormaliSE 2022), pages 58–68, May 2022.

(C15) Allison Sullivan. Hawkeye: User-Guided Enumeration of Scenarios. In *Proceedings of the 32nd International Symposium on Software Reliability Engineering* (ISSRE 2021), pages 569–578, Oct. 2021. (Acceptance: 27.5%)

(C14) Tanvir Ahmed Khan, Allison Sullivan, Kaiyuan Wang. AlloyFL: A fault localization framework for Alloy. In *Proceedings of the 2021 ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering Demonstration Track* (FSE DEMO 2021), pages 1535–1539, Aug. 2021. (Acceptance: 62.5%)

(C13) Kaiyuan Wang, Allison Sullivan, Darko Marinov, Sarfraz Khurshid. Fault Localization for Declarative Models in Alloy. In *Proceedings of the 331st International Symposium on Software Reliability Engineering* (ISSRE 2020), pages 391–402, Oct. 2020. (Acceptance: 26%)

(C12) George Thompson, Allison Sullivan. ProFL: A Fault Localization Framework for Prolog. In *Proceedings of the 29th ACM SIGSOFT International Symposium on Software Testing and Analysis Demonstration Track* (ISSTA DEMO 2021), pages 561–564, July. 2020. (Acceptance: 33%)

(C11) Allison Sullivan, Darko Marinov, Sarfraz Khurshid. Solution Enumeration Abstraction - A Modeling Idiom to Enhance a Lightweight Formal Method. In *Proceedings of the 21st International Conference on Formal Engineering Methods* (ICFEM 2019), pages 336–352, Nov. 2019. (Acceptance: 30%)

(C10) Kaiyuan Wang, Allison Sullivan, Sarfraz Khurshid. ARepair: A Repair Framework for Alloy. In *Proceedings of the 41st IEEE/ACM International Conference on Software Engineering* (ICSE Demo 2019), pages 103–106, May. 2019. (Acceptance: 47%)

(C9) Kaiyuan Wang, Allison Sullivan, Darko Marinov, Sarfraz Khurshid. ASketch: A Sketching Framework for Alloy. In *Proceedings of the 2018 ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering* (FSE Demo 2018), pages 916–919, Nov. 2018. (Acceptance: 45%)

(C8) Kaiyuan Wang, Allison Sullivan, Sarfraz Khurshid. Automated Model Repair for Alloy. In *Proceedings of the 33rd International Conference on Automated Software Engineering* (ASE 2018), pages 577–588, Sept. 2018. (Acceptance: 20%)

(C7) Kaiyuan Wang, Allison Sullivan, Darko Marinov, Sarfraz Khurshid. Solver-based Sketching of Alloy Models using Test Valuations. In *Proceedings of the 6th International Conference on Abstract State Machines, Alloy, B, TLA, VDM, and Z* (ABZ 2018), pages 121–136, June 2018. (Acceptance: 39%)

(C6) Kaiyuan Wang, Allison Sullivan, Manos Koukoutos, Darko Marinov, Sarfraz Khurshid. Systematic Generation of Non-equivalent Expressions for Relational Algebra. In *Proceedings of the 6th International Conference on Abstract State Machines, Alloy, B, TLA, VDM, and Z* (ABZ 2018), pages 105–120, June 2018. (Acceptance: 39%)

(C5) Kaiyuan Wang, Allison Sullivan, Sarfraz Khurshid. MuAlloy: A Mutation Testing Framework for Alloy. In *Proceedings of the 40th International Conference on Software Engineering Demonstration Track* (ICSE Demo 2018), pages 29–32, May 2018. (Acceptance: 42%)

(C4) Allison Sullivan, Kaiyuan Wang, Sarfraz Khurshid. AUnit: A Test Automation Tool for Alloy. In *Proceedings of the 11th IEEE International Conference on Software Testing, Verification and Validation Demonstration Track* (ICST Demo 2018), pages 398–403, Apr. 2018.

(C3) Allison Sullivan, Kaiyuan Wang, Razieh Nokhbeh Zaeem, Sarfraz Khurshid. Automated Test Generation and Mutation Testing for Alloy. In *Proceedings of the 10th IEEE International Conference on Software Testing, Verification and Validation* (ICST 2017), pages 264–275, Mar. 2017. (Acceptance: 27%)

(C2) Nima Dini, Allison Sullivan, Milos Gligoric, Gregg Rothermel. The Effect of Test Suite Type on Regression Test Selection. In *Proceedings of the 27th IEEE International Symposium on Software Reliability Engineering* (ISSRE 2016), pages 47–58, Oct. 2016. (Acceptance: 35%)

(C1) Allison Sullivan, Razieh Nokhbeh Zaeem, Sarfraz Khurshid, Darko Marinov. Towards a Test Automation Framework for Alloy. In *Proceedings of the 2014 International Symposium on Model Checking of Software* (SPIN 2014), pages 113–116, July. 2014. (Acceptance: 63%)

REFEREED WORKSHOP PUBLICATIONS

(W3) Sanyogita Piya, Anahita Samadi, and Allison Sullivan. Is More or Less Automation Better? An Investigation into the LLM4TDD Process. In *The Second International Workshop on Large Language Models for Code* (LLM4CODE@ICSE 2025), pages 161–168, June 2025.

(W2) Sanyogita Piya and Allison Sullivan. LLM4TDD: Best Practices for Test Driven Development Using Large Language Models. In *The First International Workshop on Large Language Models for Code* (LLM4CODE@ICSE 2024), pages 14–21, April 2024.

(W1) Allison Sullivan, Kaiyuan Wang, Sarfraz Khurshid, Darko Marinov. Evaluating state modeling techniques in Alloy. In *Proceedings of the 6th Workshop on Software Quality Analysis, Monitoring, Improvement, and Applications* (SQAMIA 2017), pages 16:1–16:9, Sept. 2017.

THESIS AND DISSERTATION

(T2) Allison Sullivan. Automated testing and sketching of Alloy models. In *Texas ScholarWorks - UT Electronic Theses and Dissertations*, May 2017.

(T1) Allison Sullivan. AUnit: A testing framework for Alloy. In *Texas ScholarWorks - UT Electronic Theses and Dissertations*, May 2014.

FACULTY SUPERVISOR

I have had the privilege of working with 6 amazing PhD students, 3 impressive masters students and 14 spectacular undergraduate students.

Current Students:
5 PhD students, **3** Undergraduate student

PhD (exp. Spring 2027)	**Guanxuan Wu**, Focus: Software Engineering, Formal Methods & Machine Learning
PhD (exp. Spring 2028)	**Mohammad Patwary**, Focus: Software Engineering, Formal Methods & Human Computer Interaction
PhD (exp. Spring 2031)	**Sanyogita Piya**, Focus: Software Engineering, Formal Methods & Human Computer Interaction
PhD (exp. Spring 2031)	**Jaid Chowdhury**, Focus: Software Engineering, Formal Methods & Human Computer Interaction
PhD (exp. Spring 2031)	**Hiep Nguyen**, Focus: Software Engineering, Formal Methods & Machine Learning PhD student at TCU, Co-Advised by Bingyang Wei
BS (exp. Spring 2029)	**Andy Khang Ta**, Project: Block-Based Editor for Finite Model Finders (UR2PHD Summer REU)

Graduated Students
2 PhD students, **3** M.S. students, **2** undergraduate student, **9** single semester REUs

PhD 2025	**Anahita Samadi,** Co-Advised with Jacob Luber Dissertation: "*Advancing Machine Learning Approaches Through Robust Methodologies in LLM Code Generation, Adversarial Text Classification, and Unsupervised Learning*" First job: Post-Doc, UT Southwestern
PhD 2024	**Ana Jovanovic** Dissertation: "*Improving the Accuracy of Software Models Using Refinement and Mutation Testing*" First job: Microsoft (Belgrade, Serbia)
BS. Exp. 2027	**Tiarra Marcee Payne**, REU Fall 2025 - Summer 2026 UR2Phd Academic Year REU (Verification of Geometric Map Translations) NSF REU Supplement (Live Programming for SMT Solver Prototypes)
BS. 2024	**Sanyogita Piya**, REU Summer 2023 - Spring 2024 LLM4Code@ICSE 2024 & 2025 first author paper First job: Mouser Electronics
BS. 2023	**Adam Emerson**, REU Summer 2022 - Spring 2023 ISSRE 2023 first author paper First job: Ph.D. student at UTA under Dr. Torres
MS 2020	**Jasmine Mabrey**, Thesis: *Automated Defect Classification using Machine Learning* (NC A&T) Rising Scholar Award First Job: Sandia National Laboratory
MS 2020	**George Thompson**, Thesis: *Towards Automated Fault Localization for Prolog Models* (NC A&T) Nominated for Outstanding Thesis Award First Job: Sandia National Laboratory
MS. 2018	**Regina Bunch**, Project: *Improving GUIs for the Autistic Community* (NC A&T) First Job: Cigna

1 Semester REU Students

Summer 2026	**Luke Siharath**, Project: Live Programming for SMT Solver Prototypes (NSF REU Supplement)
Spring 2026	**Joel PamPam**, Project: Verification of Geometric Map Translations (CoE REU)
Spring 2026	**Mushtahir Kabir**, Project: Block-Based Editor for Finite Model Finders (UR2PHD)
Fall 2025	**Justine Talia Gacho**, Project: Verification of Geometric Map Translations (UR2PHD, CoE REU)
Fall 2024	**Neha Joshi**, Project: Completion Rules for Alloy Set Operators (CAHSI)
Spring 2024	**Moyses Ledezma Bernal**, Project: Hint Generation for Alloy Set Operators (CAHSI)
Fall 2022	**Cristian Munoz**, Project: Generating Core Software Model from Scenarios (CAHSI)
Fall 2021	**Resha Adhikari**, Project: Empirical Study of Program Repair Techniques for Java (UTA UROP)
Spring 2020	**William Dunston**, Project: Fault Classification for CodeHints (NC A&T)
Fall 2019	**Joelle Banks**, Project: Empirical Study of Program Repair Techniques for C (NC A&T)

UNIVERSITY SERVICE

Student Committee Member
I have had the honor of serving on the following student committees:

Ph.D.	Fahmid Al Rifat (UTA), Adly Noore (UTA), Mohammad Rifat Arefin (UTA), Adam Emerson (UTA), Saif Uddin Mahmud (UTA), Pujan Budhathoki (UTA), Arjun Dahal (UTA), Sabrina Haque (UTA), Krishna Khadka (UTA), Shovon Pereira (UTA), Fadul Sikder (UTA), Shreyosi Endow (UTA), Qiping Wei (UTA), Hanan Alshahr (Fall 2019, NC A&T)
M.S. Thesis	Derrick Leflore (Fall 2019, NC A&T), Shannon Wiggins Brown (Spring 2020, NC A&T), Yuanrui Ren (Fall 2018, The University of Texas at Austin).

University Committees

Ongoing	College of Engineering Dean Search Committee

College of Engineering Committees

Ongoing	Undergraduate Curriculum Committee Member since August 2026

Departmental Committees

Ongoing	Broadening Participation in Computing (BPC) Committee Member since August 2020
Ongoing	Undergraduate Curriculum Committee Member since October 2024
Ongoing	Undergraduate Academic Program Directors Committee Member since September 2025
Ongoing	ABET Committee Member since September 2025
Ongoing	Student Engagement, Organizations and Experiences Committee Member since September 2025
2020 - 2025	PhD Admissions Committee Member
2024 - 2025	Faculty Search Committee Member
2023 - 2025	Department Student Award Committee Member
2023 - 2025	New Faculty/Instructors Orientation Committee Member
2022 - 2023	Faculty Search Sub-Chair for Software Engineering and Security
2021 - 2022	Bylaws Committee Member
2020 - 2021	Faculty Search Committee Member

Outreach Activities

Ongoing	CRA's UR2PhD Faculty Mentor since Summer 2025
Ongoing	CAHSI REU Faculty Mentor since Fall 2022
Ongoing	Faculty advisor for UTA's Society of Women Engineers since Aug. 2020
Ongoing	UTA CSE Broadening Participation in Computing Committee Member since Aug. 2020
2025	Panelist at Dallas College SWE's Women in Engineering Panel
2025	Co-organizer for the PhD Lightening Talks, Poster & Demo Session, and OurCS@DFW presentations at SCRF
2024	Judge for HackUTA
2024	Hosted a workshop on STEM majors at Digital Divas
2024	Moderator for the PhD Lightening Talks at SCRF
2023	Panelist for "Blazing Bright: Faculty Impact To The Computing" at UTA's CSE's 50th anniversary event.
2023	Invited speaker at "Never Work in Theory" industry conference
2022	Escorted a cohort of UTA students to the Grace Hopper Celebration.
2022	Hosted a workshop on Z3 at OurCS@DFW by UTA.
2022	Moderator for the "Career Paths in Computing" panel at SCRF.
2022	Panelist for "The Graduate School Experience" panel at OurCS@DFW.
2022	Hosted a workshop on STEM majors at Digital Divas
2021	Hosted a workshop on Z3 at OurCS@DFW by UTA.
2021	Panelist for the "Career Paths in Computing" panel at SCRF.
2021	Hosted a workshop on STEM majors at Digital Divas
2021	Staffed UTA's virtual booths at TAPIA and CMD-IT Graduate Fair.
2020	Organizer for the Triad Programming Contest
2020	Faculty Mentor for NC A&T's SAE International AutoDrive Challenge Student Team (Functional Safety Group)
2020	Coach for NC A&T's Advancing Minorities' Interest in Engineering (AMIE) Design Challenge team
2019	Organizer for the Triad Programming Contest
2019	Coach for NC A&T's 2019 Black Enterprise 2019 BE SMART Hackathon team. **Placed 1st**
2019	Coach for NC A&T's Advancing Minorities' Interest in Engineering (AMIE) Design Challenge team

Professional Membership

Ongoing	Society of Women Engineers
Ongoing	Association for Computing Machinery
2018 – 2020	NC A&T's Center of Excellence in Cybersecurity Research, Education and Outreach

Professional Development

2024	Affinity Research Groups: Creating and Maintaining Effective Research Groups Course
2020	Attended "Engaging Students in Online Synchronous Class Meetings"
2020	Reviewer for Google's "Technical Writing One/Two" courses.
2019	Member of the NC A&T team to recruit Facebook's Engineer in Residence Program.
2019	NC A&T's CS Representative to Intel's HBCU Consortium and Intel's HBCU Beyond 2020 grant.
2018	NC A&T Representative to Facebook T3 "Train the Trainer" summit on Data Structures and Algorithms.

PROFESSIONAL SERVICE

Organizer

Organizing Committee	The 30th International Conference on Model-Driven Engineering Languages and System (MODELS 2027) Posters Co-Chair
Organizing Committee	The 38th IEEE International Symposium on Software Reliability Engineering (ISSRE 2027) Sustainability Co-Chair
Organizing Committee	The 37th IEEE International Symposium on Software Reliability Engineering (ISSRE 2026) Publicity Co-Chair
Organizing Committee	The 36th IEEE International Symposium on Software Reliability Engineering (ISSRE 2025) Workshop Co-Chair
Organizer	"Specification Engineering: Foundations for the Future of Software Development" Dagstuhl Seminar

Conferences

PC Member	The 38th International Symposium on Software Reliability Engineering (ISSRE 2027)
PC Member	The 20th IEEE International Conference on Software Testing, Verification and Validation (ICST 2027)
PC Member	The 14th International Conference on Formal Methods in Software Engineering (FormaliSE 2027)
PC Member	The ACM International Conference on the Foundations of Software Engineering (FSE 2027)
PC Member	The 37th International Symposium on Software Reliability Engineering (ISSRE 2026)
PC Member	The 29th International Conference on Model-Driven Engineering Languages and Systems (MODELS 2026)
PC Member	The 41st International Conference on Automated Software Engineering (ASE 2026)
PC Member	The 35th International Symposium on Software Testing and Analysis (ISSTA 2026)
PC Member	The 13th International Conference on Formal Methods in Software Engineering (FormaliSE 2026)
PC Member	The 36th International Symposium on Software Reliability Engineering (ISSRE 2025)
PC Member	The 28th International Conference on Model-Driven Engineering Languages and Systems Workshops (WORKSHOPS at MODELS 2025)
PC Member	The 40th International Conference on Automated Software Engineering (ASE 2025)
PC Member	The 13th International Conference on Formal Methods in Software Engineering (FormaliSE 2025)
PC Member	The 47th International Conference on Software Engineering (ICSE 2025)
PC Member	The 35th International Symposium on Software Reliability Engineering (ISSRE 2024)
PC Member	The 17th International Conference on Software Testing, Verification and Validation Testing Tool Track (ICST Demo 2024)
PC Member	The ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering Tools and Demonstration Track (FSE Demo 2024)
PC Member	The 12th International Conference on Formal Methods in Software Engineering (FormaliSE 2024)
PC Member	The 34th International Symposium on Software Reliability Engineering (ISSRE 2023) - Fast Abstracts
PC Member	The 23rd International Working Conference on Source Code Analysis and Manipulation (SCAM 2023)
PC Member	The Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering Tools and Demonstration Track (FSE Demo 2023)
PC Member	The 16th International Conference on Software Testing, Verification and Validation Testing Tool Track (ICST 2023)
Session Chair	The 30th International Conference on Software Analysis, Evolution and Reengineering (SANER)– Program Analysis I Session
PC Member	The 30th International Conference on Software Analysis, Evolution and Reengineering (SANER 2023)
PC Member	The Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering Tools and Demonstration Track (FSE Demo 2022)
PC Member	The 33rd International Symposium on Software Reliability Engineering (ISSRE 2022)
Session Chair	The 44th International Conference on Software Engineering (ICSE) – Validation and Verification Session
PC Member	The 44th International Conference on Software Engineering Tools and Demonstration Track (ICSE Demo 2022)
PC Member	The Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering Tools and Demonstration Track (FSE Demo 2020)
PC Member	The 27th International SPIN Symposium on Model Checking of Software (SPIN 2020)
PC Member	The Special Interest Group on Computer Science Education Technical Symposium (SIGCSE 2020)
PC Member	The 5th International Conference on Research in Equity and Sustained Participation in Engineering, Computing, and Technology (RESPECT 2020)

Journals

Reviewer	Empirical Software Engineering (Impact Factor: 3.5)
Reviewer	IEEE Transactions on Software Engineering (Impact Factor: 3.33)
Reviewer	ACM Transactions on Software Engineering and Methodology (Impact Factor: 2.674)
Reviewer	Software and Systems Modeling (Impact Factor: 1.910)
Reviewer	Software Tools for Technology Transfer (Impact Factor: 1.09)

Grant Panels

Panelist	2 panels at the National Science Foundation, Virtual in 2026
Panelist	4 panels at the National Science Foundation, Virtual in 2025
Panelist	1 panel at the United States-Israel Binational Science Foundation, Virtual in 2025
Panelist	1 panel at the National Science Foundation, Virtual in 2024
Panelist	2 panel at the National Science Foundation, Virtual in 2023
Panelist	1 panel for the Department of Energy in 2022
Panelist	1 panel at the National Science Foundation, Virtual in 2022
Panelist	1 panel for the Department of Energy in 2021
Panelist	2 panels at the National Science Foundation, Alexandria, VA in 2020

Other

Participant	Clarivate's 15th Academic Reputation Survey
Speaker	NSF CISE CAREER Workshop 2024
Speaker	Never Work in Theory - A Bridge Between Researchers and Practitioners.
Co-Reviewer	The 27th International Symposium on Software Testing and Analysis (ISSTA 2018)
Co-Reviewer	The 11th International Conference on Software Testing, Verification and Validation (ICST 2018)
Co-Reviewer	The 25th International Symposium on the Foundations of Software Engineering (FSE 2017)
Co-Reviewer	The 10th International Conference on Software Testing, Verification and Validation (ICST 2017)
Co-Reviewer	The 31st IEEE/ACM International Conference on Automated Software Engineering (ASE 2016)

CLASSROOM TEACHING

The University of Texas at Arlington

Fall 2026	HONR 2106: Research Training for College of Engineering Undergrads, Undergraduate
Fall 2026	CSE 4322: Software Project Management, Undergraduate
Fall 2025	CSE 3318: Data Structures and Algorithms, Undergraduate
Fall 2024	CSE 3318: Data Structures and Algorithms, Undergraduate
Spring 2024	CSE 4321: Software Testing & Maintenance, Undergraduate
Fall 2023	CSE 4321: Software Testing & Maintenance, Undergraduate
Fall 2022	CSE 4321: Software Testing & Maintenance, Undergraduate
Spring 2022	CSE 5320/6392: Computer-Aided Verification for Systems, Graduate
Fall 2021	CSE 4321: Software Testing & Maintenance, Undergraduate
Spring 2021	CSE 5320/6392: Computer-Aided Verification for Systems, Graduate
Fall 2020	CSE 5311: Design and Analysis of Algorithms, Graduate

North Carolina A&T State University

Spring 2020	COMP 496: Senior Design, Undergraduate
Spring 2020	COMP 681: Formal Methods, Graduate
Fall 2019	COMP 285: Analysis of Algorithms, Undergraduate
Fall 2019	COMP 611: System Testing and Evaluation, Graduate
Spring 2019	COMP 496: Senior Design, Undergraduate
Spring 2019	COMP 681: Formal Methods, Graduate
Fall 2008	COMP 285: Analysis of Algorithms, Undergraduate